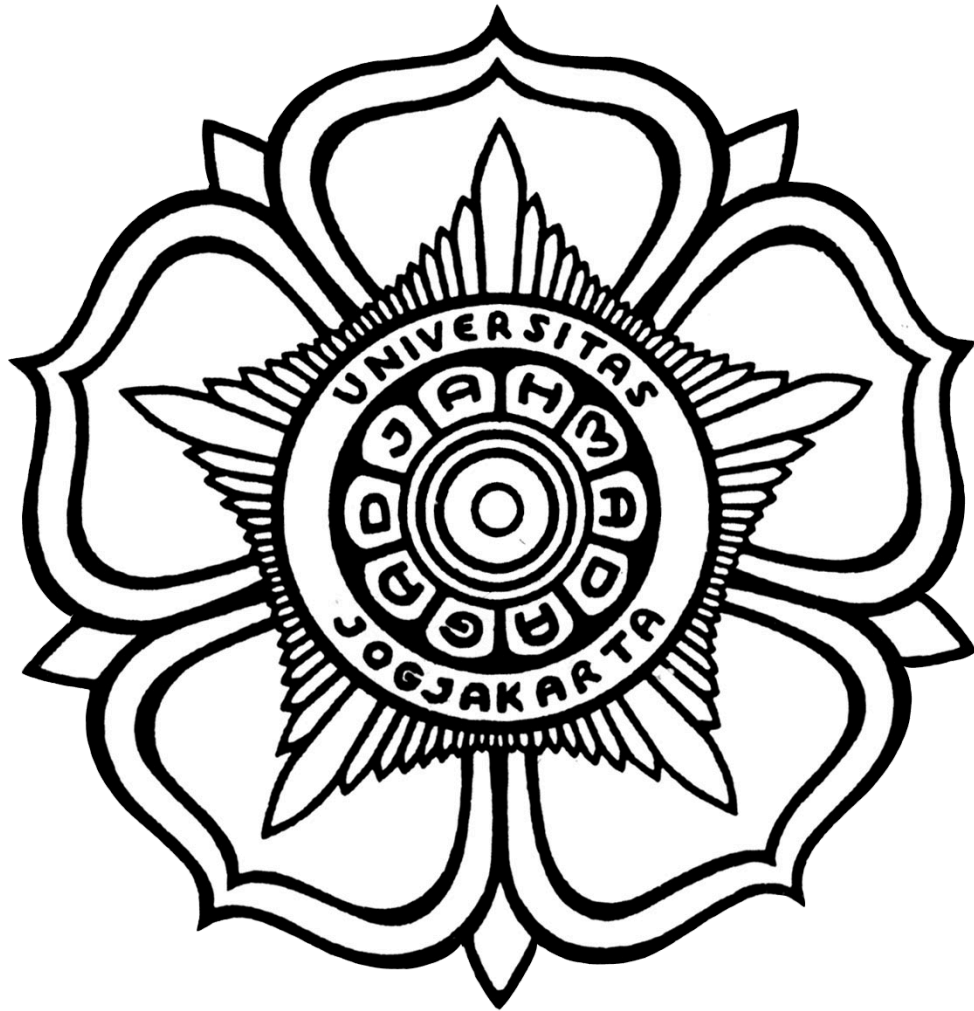


ALGORITHMIC TRADING PROJECT REPORT



INDUSTRIAL-GRADE ALGORITHMIC TRADING EA

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1. Project Background

Algorithmic trading is a method of using programmed rules to automatically analyze the market and execute trades. In MetaTrader 5, an Expert Advisor (EA) can be developed to perform these tasks based on a specific trading strategy.

This project focuses on developing an industrial-grade Expert Advisor for MetaTrader 5. The EA is designed to identify market trends, enter trades during pullbacks, and manage trading risk automatically. The strategy is intended to be systematic so that the same rules can be applied consistently without manual decisions for every trade.

The project follows several requirements, including avoiding martingale, grid trading, and high-frequency trading. The EA also uses controlled risk per trade and a monthly loss limit. Backtesting and optimization are performed using Exness market data through MetaTrader 5.

The main targets of the project are approximately 3%–5% monthly return, 50%–70% annual return, and a maximum drawdown of approximately 25%–30%. These values are treated as project targets for evaluation rather than guaranteed results.

Repository

The source code for the Industrial-Grade Algorithmic Trading EA Project

GitHub repository: <https://github.com/nailasalmayusroini/Sains-Manajemen-Project2>

2. EA Overview

The developed EA is called IndustrialTrendPullbackEA. It is implemented in MQL5 and tested using the MetaTrader 5 Strategy Tester.

The EA uses a combination of three main indicators:

- EMA (Exponential Moving Average) – identifies the general market trend.
- RSI (Relative Strength Index) – identifies pullback conditions and helps confirm entries.
- ATR (Average True Range) – measures market volatility and is used to calculate dynamic stop-loss and take-profit levels.

The strategy uses an H1 timeframe. Instead of opening trades continuously on every tick, the entry conditions are checked when a new H1 candle is formed. This helps keep the strategy away from high-frequency trading behavior.

The EA also includes several risk management features. The default risk per trade is 0.5% of account equity, only one position is allowed per symbol, and a monthly loss limit is used to stop new entries after a specified loss level is reached.

Parameter	Value
Timeframe	H1
Fast EMA	50
Slow EMA	300
RSI Period	14
RSI Buy Level	46
RSI Sell Level	56
ATR Period	14
Risk per Trade	0.50%
Stop Loss	$1.75 \times \text{ATR}$
Take Profit	$2.50 \times \text{ATR}$
Maximum Monthly Loss	5%
Maximum Positions per Symbol	1
Break-Even	Enabled
Trailing Stop	Enabled

The EA does not use martingale, grid trading, or HFT.

3. Strategy Evaluation Report

a. Trend Identification

The EA first identifies the overall market direction using two EMA indicators.

For a buy trade:

- Fast EMA must be above Slow EMA.
- Price must be above the Fast EMA.
- RSI must show a recovery from a lower level.

For a sell trade:

- Fast EMA must be below Slow EMA.
- Price must be below the Fast EMA.
- RSI must show a decline from a higher level.

The purpose of the EMA conditions is to avoid entering against the main market trend.

b. Pullback Confirmation

RSI is used to confirm that the market has experienced a pullback before entering a trade.

The optimized RSI levels are:

- Buy level: 46
- Sell level: 56
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For example, a buy signal can occur when RSI was previously below 46 and then moves back above 46 while the larger trend is bullish. This approach attempts to enter after a temporary price movement against the main trend instead of entering randomly.

c. Stop Loss and Take Profit

ATR is used to adjust the trading distances based on market volatility.

The selected values are:

- Stop Loss = $1.75 \times \text{ATR}$
- Take Profit = $2.50 \times \text{ATR}$

Using ATR means that the stop-loss and take-profit distances are not fixed numbers. They change according to the current volatility of the instrument.

d. Risk Management

The EA uses several risk management features:

1. 0.5% risk per trade
2. Maximum one position per symbol
3. Monthly loss limit of 5%
4. Break-even protection
5. ATR trailing stop
6. No martingale
7. No grid
8. No HFT

These features are intended to prevent excessive exposure and reduce the effect of individual losing trades.

4. Optimization and Backtest Setup

The strategy was optimized using the MetaTrader 5 Strategy Tester. The optimization focused on the main strategy parameters, including EMA periods, RSI levels, and ATR multipliers. The optimization used Profit Factor as the main optimization criterion. The parameter search was kept relatively limited to reduce the possibility of overfitting.

The final parameter combination selected for further testing was:

- Fast EMA = 50

- Slow EMA = 300
- RSI Buy Level = 46
- RSI Sell Level = 56
- SL ATR Multiplier = 1.75
- TP ATR Multiplier = 2.50

After optimization, the selected parameters were tested on instruments that were not used as the optimization selection period.

Backtest Settings

Setting	Value
Platform	MetaTrader 5
Broker	Exness
Initial Balance	USD 10,000
Leverage	1:1
Timeframe	H1
Model	Every tick based on real ticks
Optimization	Disabled during OOS tests
OOS Period	24 September 2023 – 23 September 2026

The out-of-sample tests cover approximately three years. The broader project setup uses a seven-year historical window, but the results presented below should not be described as seven-year out-of-sample results.

5. Instruments Tested

The EA was tested across different instruments available through the Exness MT5 environment.

The completed valid tests used the following ten instruments:

- EURUSD
- GBPUSD
- USDJPY
- USDCHF
- USDCAD
- NZDUSD
- USDCNH
- US500
- USTEC

- JPN225

XAUUSD was also tested, but its result was not included in the final comparison because the test only had 44% history quality and zero trades. Therefore, the XAUUSD result was considered inconclusive rather than a real 0% performance result.

6. Strategy Evaluation Report

The following table shows the out-of-sample backtest results using the same selected parameters.

Instrument	Type	History	Net Profit	Return	Max Equity DD	Profit Factor	Trades	Wins	Losses
EURUSD	Forex	100%	-\$78.44	-0.78%	1.24%	0.46	6	3	3
GBPUSD	Forex	99%	-\$45.68	-0.46%	0.69%	0.00	2	0	2
USDJPY	Forex	99%	-\$1.31	-0.01%	1.39%	0.99	11	5	6
USDCHF	Forex	99%	+\$128.89	+1.29%	0.95%	3.59	8	5	3
USDCAD	Forex	99%	+\$89.56	+0.90%	0.69%	2.80	2	1	1
NZDUSD	Forex	99%	-\$27.84	-0.28%	1.21%	0.81	8	5	3
USDCNH	Forex	99%	\$0.00	0.00%	0.00%	N/A	0	0	0
US500	Index	99%	-\$282.77	-2.83%	9.73%	0.95	246	125	121
USTEC	Index	99%	-\$2,419.59	-24.20%	25.92%	0.59	259	115	144
JPN225	Index	99%	-\$2,995.52	-29.96%	40.41%	0.84	246	111	135

7. Result Analysis

The backtest results show that the performance of the EA was different for each instrument. Among the tested instruments, USDCHF produced a positive return of 1.29%, with a profit factor of 3.59 and a maximum equity drawdown of only 0.95%. USDCAD also produced a positive return of 0.90%, with a profit factor of 2.80 and a maximum equity drawdown of 0.69%.

USDJPY was almost at break-even, with a return of -0.01% and a profit factor of 0.99. This means that the total profits and losses were relatively close during the test. Several Forex instruments produced relatively small losses. EURUSD returned -0.78%, GBPUSD returned -0.46%, and NZDUSD returned -0.28%.

US500 produced a larger loss of -2.83%, but its maximum equity drawdown was 9.73%, which remained below the project's 25–30% drawdown target. The largest negative results came from

the other two index instruments. USTEC produced a return of -24.20% with a maximum equity drawdown of 25.92%. JPN225 produced a return of -29.96% and a maximum equity drawdown of 40.41%.

The JPN225 result is particularly important because its maximum drawdown was higher than the project's maximum drawdown target.

8. Comparative Summary

The results can be grouped into several general categories.

Positive Results

Instrument	Return	Profit Factor	Max DD
USDCHE	+1.29%	3.59	0.95%
USDCAD	+0.90%	2.80	0.69%

These two instruments were the only instruments that produced positive returns during the tested out-of-sample period.

Close to Break-Even

Instrument	Return	Profit Factor
USDJPY	-0.01%	0.99
USDCNH	0.00%	N/A

USDCNH should not be considered profitable because it had **zero trades**. Its result only means that the strategy did not open a position during the test period.

Moderate Losses

Instrument	Return	Max DD
EURUSD	-0.78%	1.24%
GBPUSD	-0.46%	0.69%
NZDUSD	-0.28%	1.21%
US500	-2.83%	9.73%

These instruments produced losses, but their drawdowns were considerably lower than the large index losses.

Large Losses

Instrument	Return	Max DD
USTEC	-24.20%	25.92%
JPN225	-29.96%	40.41%

The results on USTEC and JPN225 show that the same strategy does not perform equally well across different markets.

9. Comparison with Project Targets

The project has several performance targets. The actual results are compared with these targets below.

Requirement	Result
Monthly return around 3%–5%	Not consistently achieved
Annual return around 50%–70%	Not achieved in the reported OOS results
Maximum DD around 25%–30%	Mixed; JPN225 reached 40.41%
No more than six losing months in a year	Requires separate monthly analysis
No martingale	Achieved
No grid	Achieved
No HFT	Achieved
Every tick based on real tick	Achieved for the reported tests
1:1 leverage in tester	Achieved

The results therefore do not show that the current strategy meets all of the requested performance targets. However, the EA structure follows the requested trading restrictions and includes several risk management features.

The return targets should not be achieved by simply increasing the risk per trade, because doing so could also increase drawdown. The current results are therefore kept as the actual backtest results.

10. Strengths and Weakness of the EA

Strengths

The EA has several useful features for an automated trading project:

- Uses clear rule-based entry conditions.
- Combines trend and pullback confirmation.

- Uses ATR for dynamic risk levels.
- Uses percentage-based position sizing.
- Includes break-even and trailing-stop functions.
- Has a monthly loss protection mechanism.
- Limits positions per symbol.
- Does not use martingale or grid trading.
- Uses new-bar entry logic instead of continuously opening trades.

Weaknesses

The backtest also shows several weaknesses:

- Performance varies significantly between instruments.
- The strategy does not consistently reach the required return targets.
- Some instruments produced negative profit factors.
- JPN225 exceeded the maximum drawdown target.
- Some tests had very few trades, making their results less reliable.
- USDCNH produced no trades during the test period.
- The current portfolio-risk input needs to be properly implemented or removed from the final version of the code.

11. Conclusion

The project successfully developed and tested an automated trading EA for MetaTrader 5 using a trend-following pullback strategy. The EA combines EMA, RSI, and ATR indicators with several risk management features, including percentage-based risk, break-even protection, trailing stops, and a monthly loss limit.

The optimized parameter set used Fast EMA 50, Slow EMA 300, RSI buy level 46, RSI sell level 56, SL ATR multiplier 1.75, and TP ATR multiplier 2.50. The strategy was tested on ten instruments using H1 data, real ticks, a USD 10,000 initial balance, and 1:1 leverage.

The results showed that the strategy performed differently depending on the instrument. USDCHF and USDCAD produced positive returns, while USDJPY was close to break-even. Other instruments produced losses, with JPN225 and USTEC showing the largest negative results. JPN225 also exceeded the project's maximum drawdown target.

Overall, the backtest shows that the EA follows the main technical requirements of the project, but its current performance does not consistently meet the requested return targets across all instruments. Further improvement could focus on making the strategy more stable across different market types, testing the strategy over the complete seven-year period with fixed parameters, and improving the portfolio-level risk control.

The project demonstrates the complete process of developing, optimizing, testing, and evaluating an automated trading strategy while keeping the actual results transparent.